

\* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

\* Rank 1 tensor.

$$[0 \quad 0 \quad Q_z]$$

$$Q_z = \mathbb{Q}_z^{(1)}$$

\* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{yy} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

$$Q_{yy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$

$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

\* Rank 2 tensor (a). \* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{yyz} \\ 0 & 0 & Q_{zzz} \\ 0 & Q_{yyz} & 0 \\ Q_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\beta(1)}$$

$$Q_{yyz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\beta(1)}$$

$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

\* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{yyz} \\ 0 & 0 & 0 \\ 0 & -\frac{Q_{yyz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)}$$

$$Q_{yyz} = -2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)}$$

$$\frac{Q_{yyz}}{2} = -\mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)}$$

$$\frac{Q_{xxz}}{2} = \mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)}$$

\* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & Q_{yzy} & 0 \\ Q_{zxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzy} = \mathbb{G}_{xy}^{(2)} + \mathbb{Q}_z^{(3)}$$

$$Q_{zxx} = \mathbb{G}_{xy}^{(2)} - \mathbb{Q}_z^{(3)}$$

\* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & 0 & 0 & 0 \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$

$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$

$$Q_{zxxz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)}$$

\* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 \\ -Q_{xxyy} & 0 & Q_{yyzz} & 0 & 0 & 0 \\ -Q_{xxzz} & -Q_{yyzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} - 2\mathbb{Q}_v^{(3)}$$

$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(3)}$$

$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(3)}$$

\* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{zxxz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(4)}$$

$$Q_{zxxz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 2\mathbb{Q}_v^{(4)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

\* Rank 4 tensor (aaa). \* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yzyz} & 0 & 0 \\ 0 & Q_{zxxz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)}$$

$$Q_{xyxy} = \mathbb{G}_{xyz}^{(2)} + 2\mathbb{Q}_v^{(5)}$$

\* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(3)} - 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(3)} + 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)}$$

$$Q_{xyxy} = \mathbb{G}_{xyz}^{(3)} + 2\mathbb{Q}_v^{(6)}$$

\* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyyx} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$

$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{yyxx} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} + 2\mathbb{Q}_v^{(3)}$$

$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$

$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{zzxx} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{zzyy} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{yzzz} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{zxzx} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

$$Q_{xyxy} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(5)}$$

$$Q_{xyyx} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(5)}$$

\* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyyx} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)}$$

$$Q_{yzzz} = \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)}$$

$$Q_{xyxy} = \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)}$$

$$Q_{xyyx} = \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} - 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)}$$

\* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & 0 \\ 0 & Q_{yyyy} & 0 \\ 0 & 0 & Q_{zzzz} \\ 0 & 0 & Q_{yyzz} \\ Q_{zzxx} & 0 & 0 \\ 0 & Q_{xxyy} & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & 0 & Q_{zzxx} \\ Q_{xxyy} + Q_{yyzz} - Q_{yyzz} - Q_{zzxx} + Q_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3Q_0^{(1)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} + 2Q_4^{(1)} - 2Q_u^{(1)} - 4Q_u^{(2)} + 2Q_v^{(1)} + 4Q_v^{(2)}$$

$$Q_{yyyy} = 3Q_0^{(1)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} + 2Q_4^{(1)} - 2Q_u^{(1)} - 4Q_u^{(2)} - 2Q_v^{(1)} - 4Q_v^{(2)}$$

$$Q_{zzzz} = 3Q_0^{(1)} + 2Q_{4u}^{(1)} + 2Q_4^{(1)} + 4Q_u^{(1)} + 8Q_u^{(2)}$$

$$Q_{yyzz} = Q_0^{(1)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(1)} + 3Q_u^{(3)} - Q_v^{(1)} + Q_v^{(3)}$$

$$Q_{zzxx} = Q_0^{(1)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(1)} - 3Q_u^{(3)} + Q_v^{(1)} + Q_v^{(3)}$$

$$Q_{xxyy} = Q_0^{(1)} + 2Q_{4u}^{(1)} - Q_4^{(1)} - 2Q_u^{(1)} - 2Q_v^{(3)}$$

$$Q_{yyzz} = Q_0^{(1)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(2)} + 3Q_u^{(5)} - Q_v^{(2)} + Q_v^{(5)}$$

$$Q_{zzxx} = Q_0^{(1)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(2)} - 3Q_u^{(5)} + Q_v^{(2)} + Q_v^{(5)}$$